PHASE 1: OFFLINE TRAINING

Training Partition $\mathcal{D}_{\text{train}}$ (80%)

Stratified 80/10/10 Splits
Benchmark Cross-Evaluation

Stage 1: Primary GBDT Core

Cost-Weighted Ensemble $f_{\text{core}}(\mathbf{x})$
Balanced Weighting (w_{pos})

Validation Score

PHASE 2: VALIDATION & CALIBRATION

Validation Partition $\mathcal{D}_{\text{val}}$ (10%)

Validation Inference: $\hat{p}_{\text{val}} = f_{\text{core}}(\mathbf{x})$
Held-Out Tuning Partition

Stage 2: Optimal Threshold τ^* Search

$\tau^* = \arg\min_{\tau} \text{Cost}(\tau)$
Uniform Grid Search $\tau \in [0.01, 0.99]$

$\hat{p} \geq \tau^*$

5-Fold OOF Cross-Fitting

Candidate Alerts $\mathcal{D}_{\text{cand}}$ ($\hat{p} \geq \tau^*$)
TreeSHAP $\phi(\mathbf{x})$ + Meta-Model $g^{(j)}$

OOF Meta-Risk $\hat{q}$

Amount-Tiered Safety Calibration

Quantile Stratification ($K = 3$ Tiers)
Safety Thresholds $\tau_{\text{meta},k}$ ($\Delta\text{FN} = 0$)

PRODUCTION DEPLOYMENT PIPELINE (Algorithm 1) : Deploy $\{f_{\text{core}}, \tau^*, g^{(j)}, \{\tau_{\text{meta},k}\}\} \longrightarrow$ Streaming Runtime Engine

Deploy f_{core} & τ^*

Deploy g & $\{\tau_{\text{meta},k}\}$

PHASE 3: STREAMING PRODUCTION DEPLOYMENT WITH SAFETY GUARD

Live Transaction

Feature Vector $\mathbf{x} \in \mathbb{R}^d$
Real-Time Ingestion
Single-Row ($B = 1$)

Stage 1 Fast-Path

Score $\hat{p} = f_{\text{core}}(\mathbf{x})$
Sub-millisecond Scoring
Gate: $\hat{p} \geq \tau^*$?

$\hat{p} < \tau^*$

Auto-Cleared Legit

Direct Approval ($\hat{y} = 0$)

$\hat{p} \geq \tau^*$

Stage 3 TreeSHAP

Local Attribution $\phi(\mathbf{x})$
Exact TreeExplainer
Selective Trigger

Meta-Classification

Joint Input $[\mathbf{x}, \phi(\mathbf{x})]$
Ensemble $g(\mathbf{u}) \to \bar{q}$
Secondary Filter

Safety Guard

Tier k(Amount)
$\bar{q} \geq \tau_{\text{meta},k}$?

$\bar{q} \geq \tau_{\text{meta},k}$

Auto-Cleared False Alarm ($\hat{y} = 0$)

$\bar{q} < \tau_{\text{meta},k}$

Escalated Candidate Alert ($\hat{y} = 1$)